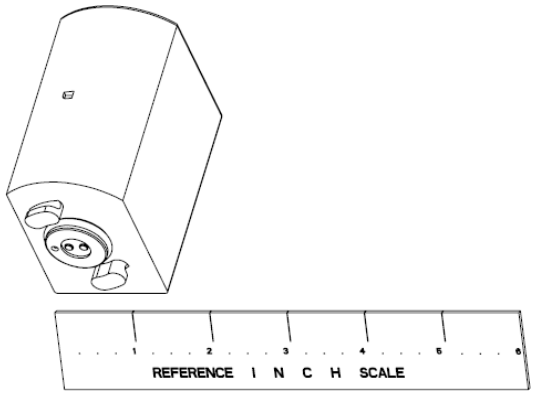
INCH-POUND
MIL-PRF-32271/10B
03 JUNE 2025
SUPERSEDING
MIL-PRF-32271/10A
26 MAY 2010

PERFORMANCE SPECIFICATION SHEET

BATTERY, NON-RECHARGEABLE, LITHIUM

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product herein shall consist of this specification sheet and MIL-PRF-32271.



See NOTES for nominal dimensions; see applicable drawing for dimensional requirements.

FIGURE 1. General view, M32271/10 battery shape

REFERENCES

NAVAIR 1794AS053 Performance Specification for Batteries, Non-Rechargeable, High Performance (LIMNO2)

(Copies of this document are available from Naval Air Systems Command, Patuxent River, MD 20670-1547)

Unless otherwise specified, paragraph numbers are in reference to MIL-PRF-32271. Note that the requirement and verification paragraphs included in this document that are in addition to or modified from MIL-PRF-32271 are highlighted in **BOLD**.

For clarity, MIL-PRF-32271 tables are labeled with capital Roman numerals (e.g. IX) and MIL-PRF-32271/10B tables are labeled with lower case Roman numerals (e.g. ix).

Distribution Statement A. Approved for public release: distribution is unlimited.
Source: https://assist.dla.mil -- Downloaded: 2026-10-21T05:38Z
Check the source to verify that this is the current version before use.

REQUIREMENTS (see 1.2 for Type, Class, and Features descriptions):

Type	I
Class	2
Features	D
PIN	M32271/10-12D

Specification requirements: The following requirements of MIL-PRF-32271, identified therein by the phrase “when specified”, are applicable as indicated below:

TABLE i. *“When specified” requirements*

Requirement	**Requirement Paragraph**	**Verification Paragraph**	**Applicability by PIN Suffix (M32271/9-)**
			12D
Parallel discharges	*Tables II, III, IV, IX, X, XI, & XII*		*No*
Additional material requirements	*3.3.5*	*4.6.1.11*	*No*
Parallel cell arrangements[1]	3.4		Yes
Connectors[2]	3.4.5	4.6.1.7[3]	Yes
Battery charger connection	*3.4.5.1*	*4.6.1.7.1*	*No*
Connection integrity	3.4.5.2		
Static	3.4.5.2	4.6.1.7.2	Yes
Dynamic	3.4.5.2	4.6.1.7.3	Yes
Terminal integrity	*3.4.6*		
Socket Strength	*3.4.6.1*	*4.6.1.8.1*	*No*
Terminal Strength	*3.4.6.2*	*4.6.1.8.2*	*No*
Complete discharge device	3.4.7	4.6.1.9	Yes
State of charge device	3.4.8	4.6.1.10	Yes
State of charge data output	*3.4.8e*	*4.6.1.10*	*No*
Additional design and construction requirements	3.4.11	4.6.1.12	Yes
Cell charging	3.5.3	4.6.2.3	Yes
Nail penetration	3.5.4	4.6.2.4	Yes
Parallel string charge protection[4]	3.5.6	4.6.2.6	Yes
Cell series string short circuit	3.5.7	4.6.2.7	Yes
Battery charge protection	3.5.12	4.6.2.12	Yes
Over-current protection	3.5.14.1	4.6.2.14	Yes
Over-temperature protection	3.5.15	4.6.2.15	Yes
Projectile	3.5.16	4.6.2.17	Yes
Additional safety requirements	*3.5.18*	*4.6.2.18*	*No*
Capacity LR	*3.6*	*4.6.4.1.5*	*No*
Capacity LRT	*3.6*	*4.6.4.1.9*	*No*

[1] Parallel cell arrangements are permitted but are not required
[2] For continuity only
[3] Verification for continuity only; insertion and withdrawal measurements not required for this battery
[4] Only required if parallel cell arrangements are used

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T05:38Z
Check the source to verify that this is the current version before use.

TABLE i. *"When Specified" Requirements* - Continued

Requirement	Requirement Paragraph	Verification Paragraph	Applicability by PIN Suffix (M32271/9-)
			12D
Capacity IDC	*3.6*	*4.6.4.1.10*	*No*
Surface temperature	3.6h	4.6.4.1	Yes
Immersion	*3.7.8*	*4.6.5.8*	*No*
Watertight integrity	*3.7.9*		
Shallow	*3.7.9*	*4.6.5.9*	*No*
Deep	*3.7.9*	*4.6.5.10*	*No*
Electromagnetic interference	3.7.10	4.6.5.11	Yes
Electrostatic discharge	3.7.11	4.6.5.12	Yes
Additional environmental requirements	3.7.12	4.6.5.13	Yes
DO NOT CHARGE! marking	*3.8.1.2b*		*No*
Complete discharge device marking/label	3.8.4		Yes
State of charge marking	3.8.5		Yes
Simplified battery label	3.8.7		Yes
Additional markings	*3.8.8*		*No*

Dimensions, marking and configuration

Battery – Drawing A3315887
Battery Connector (3.4.5 & 4.6.1.7) – Drawing A3315887
Mating Connector – General Dynamics P/N 01-P49797J001
Battery Charger Connector – Not Applicable

Battery voltages (3.4.4 & 4.6.1.5):

Battery open-circuit voltage (3.4.4.1 & 4.6.1.5.1):
Maximum – 13.2 volts
Minimum – 12.0 volts

Battery closed circuit voltage (3.4.4.2 & 4.6.1.5.2):
For capacity and initial voltage delay:
7.0 volts for H2, L & LT tests
7.5 volts for all other capacity tests.

For battery closed circuit voltage test:
Minimum – 8.0 volts

Cell closed circuit voltage test (3.4.4.3 & 4.6.1.3):

Minimum - 2.0 volts per cell

Maximum weight (3.1 & 4.6.1.6):

0.6 lbs. (272 g)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T05:38Z
Check the source to verify that this is the current version before use.

Connection integrity, static and dynamic (3.4.5.2; 4.6.1.7.2; & 4.6.1.7.3): When subjected to the static or dynamic connection integrity tests, battery voltage shall remain above 12 volts and shall meet the requirements of 3.4.5.2.

Capacity test requirements (3.6 & 4.6.4):

Test	Initial voltage delay Time (MAX) to Volts		Minimum Capacity		Cut-off Volts
			Storage Period		
			1-Week	4-Week	
I	1 second	7.5	15.0 hours		7.5
L	1 second	7.0	1.5 hours		7.0
H	1 second	7.5	17.0 hours		7.5
H2	1 second	7.0	6.5 hours		7.0
IT	1 second	7.5	13.9 hours	13.5 hours	7.5
LT	1 second	7.0	1.4 hours	1.3 hours	7.0
HT	1 second	7.5	15.8 hours	15.3 hours	7.5

Abuse test minimum capacity requirement (3.5.12 & 4.6.2.13l):

3.0 ampere-hours.

Dust Cap. Each battery as delivered shall be provided with a dust cap that covers the battery terminals. The cap shall be snug-fitting and shall be removable by hand at temperatures from –40 to 130ºF. The material used for the dust cap shall be non-toxic and non-flammable, shall withstand temperatures from –40 to 195ºF without shrinkage or cracking, and shall not affect the performance of the o-ring in providing a watertight seal against the radio. The dust cap shall not leave any residue on the battery contacts nor have any adverse affect on the battery/radio interface. The dust cap should either be black or closely match the color of the battery. Compliance to these requirements shall be certified (3.2.1).

Magnet. Each battery shall include a magnet conforming to the magnet requirements of NAVAIR 1794AS0953, to be used to identify its rate capability to HOOK-112 radios. See “VARIANCES” herein for requirements and test methods.

METHODS OF EXAMINATION AND TEST:

Cell closed-circuit voltage test: When cells are tested as specified in 4.6.1.3, load each cell to be used in a single series cell string design with 1 ampere constant current, 2 ohms constant resistance, or 2 watts constant power. In the event that parallel cell arrangements are used, use a load that produces a current equivalent to 1 ampere divided by the number of parallel cell strings to be used, as a minimum.

Battery closed-circuit voltage test: When tested as specified in 4.6.1.5.2, load the battery with 1 ampere constant current, 8 ohms constant resistance, or 8 watts constant power.

Dimensions (4.6.1.6). Reference Drawing A3315887. Dimension D6 shall be measured in not less than two places: at the top (terminal end) and bottom of the battery. The same tolerance shall apply in both locations.

Static connection integrity (4.6.1.7.2): Connect batteries subjected to this test to test fixture, General Dynamics P/N 01-P49797J001. Center applied weight at a location 2.00 ± 0.125 inches from the terminal end of the battery.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T05:38Z
Check the source to verify that this is the current version before use.

Dynamic connection integrity (4.6.1.7.3). Connect batteries subjected to this test to test fixture, General Dynamics P/N 01-P49797J001.

Abuse test, pre-discharge: When tested as specified in 4.6.2.13a, discharge with a load of either 1 ampere or 12 watts for 1.6 hours.

Abuse test, pulse discharge: When tested as specified in 4.6.2.13l, discharge with a load of 1.25 amperes for 1 minute followed by 1.1 amperes for 4 minutes, cycled continuously to 7.5 volts.

Battery over-current protection: When tested as specified in 4.6.2.14, load batteries with either 3.75 amperes or 45 watts.

Capacity tests (4.6.4):

Test	Discharge Rate	Duty Cycle
I	See Below	Continuous discharge to cut-off volts, followed by 650 milli-amperes to zero volts, then 650 milli-amperes forced discharge for 5 minutes
L	See Below	Continuous discharge to cut-off voltage
H	See Below	Continuous discharge to cut-off voltage
H2	See Below	Continuous discharge to cut-off voltage
IT	See Below	Continuous discharge to cut-off voltage
LT	See Below	Continuous discharge to cut-off voltage
HT	See Below	Continuous discharge to cut-off voltage

Discharge Rate for all capacity tests except H2 (4.6.4): Discharge continuously at 1.8W for 267 seconds, followed by 7.2W for 30 seconds, followed by 26W for 3 seconds, to cut-off volts.

Electromagnetic interference tests (4.6.5.11): The following tests shall be performed:

TEST	TEST DESCRIPTION	VALUE	FREQUENCY
RS103	Radiated Susceptibility, E-field, Survival Only	200V/M	14 KHZ – 18 GHZ
RE102	Radiated Emission, E-field	(Info only)	2 MHZ – 18 GHZ

Discharge Rate for H2 capacity test (4.6.5.13.1e): Discharge continuously at 8.5 watts for 2.1 seconds followed by 3.0 watts for 18.9 seconds to cut-off volts.

Watertight integrity, legacy (4.6.5.13.1): Connect batteries subjected to this test to a connector consisting of General Dynamics P/Ns 64-P21451J001 (coupler plate, 1 each), 84-P21427J001 (PWB, 1 each), and 39-P21428J001 (contact, 2 each), whenever connection to a test fixture is required by 4.6.5.13.1. When assembled, ensure that this fixture forms an air-tight seal against the battery. Parts and materials in addition to those specified are permissible in order to obtain the air-tight seal provided they do not extend below the contact surface of the battery.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T05:38Z
Check the source to verify that this is the current version before use.

VARIANCES: Each battery shall comply with MIL-PRF-32271 except as follows:

Add:

3.4.11.1 Magnet. Each battery shall include a magnet conforming to 3.1.1, 3.1.8, and Figure 2 of NAVAIR 1794AS0953 for purposes of identifying its rate capability to HOOK-112 radios. When tested as specified in 4.6.1.12.1, the magnet shall have not less than 15 ampere-turns of magnetomotive force.

Add:

3.7.12.1 Watertight integrity, legacy. Batteries shall retain their watertight integrity and shall be capable of safe and continuous use when subjected to extreme environmental conditions. Test samples, when subjected to any part of the Watertight integrity tests of 4.6.5.13.1, shall meet the requirements described below.

a. Altitude cycling. Each battery shall meet the visual mechanical and battery open circuit and closed circuit voltage requirements of 3.4.4 prior to and following the altitude cycling of 4.6.5.13.1a. Weight loss of batteries shall be not greater than 0.1 grams during this test. The labels used on the battery shall show no evidence of softening, peeling, or blistering after test.
b. Explosive decompression. Following the Explosive Decompression test of 4.6.5.13.1b, each battery shall meet the visual mechanical and battery voltage requirements (see TABLE XIV and 3.4.4). Weight loss of batteries shall be not greater than 0.1 grams during this test. The labels used on the battery shall show no evidence of softening, peeling, or blistering after test.
c. Parachute drop. Following the Parachute Drop test of 4.6.5.13.1c, weight increase of each battery shall be not greater than 0.1 grams and no water shall come into contact with the battery contacts. Each battery shall meet the visual mechanical and battery open circuit and closed circuit voltage requirements (see TABLE XIV and 3.4.4). The labels used on the battery shall show no evidence of softening, peeling, or blistering after test.
d. Submersion. Following the Submersion test of 4.6.5.13.1d, weight increase of each battery shall be not greater than 0.1 grams and no water shall come into contact with the battery contacts. There shall be no sign of corrosion on the battery contacts. Each battery shall meet the visual mechanical and battery open circuit and closed circuit voltage requirements (see TABLE XIV and 3.4.4). The labels used on the battery shall show no evidence of softening, peeling, or blistering after test.
e. Capacity H2. When tested in accordance with 4.6.5.13.1e, batteries shall meet the requirements of 3.6.

Add: The following to TABLE XI of MIL-PRF-32271 after “Watertight integrity”:

Inspection or Test	Requirement Paragraph	Method or Test Paragraph	Test sample break-out by test
Submersion	3.7.12.1d	4.6.5.13.1d	TABLE XIII
Capacity H2	3.7.12.1e	4.6.5.13.1e	TABLE XIII

Add: The following to TABLE XIII of MIL-PRF-32271 after “Watertight integrity”:

Features:	A			B&C			D&E		
Inspection levels:	1	2	3	1	2	3	1	2	3
TESTS									
Submersion	10	16	27	7	12	22	3	5	12
Capacity H2	10	16	27	7	12	22	3	5	12

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T05:38Z
Check the source to verify that this is the current version before use.

Add:

4.6.1.12.1 Magnet test. Each battery shall be tested for magnetomotive force within the volume specified by Figure 2 of NAVAIR 1794AS0953.

Add:

4.6.5.13.1 Watertight integrity, legacy. The following test shall be performed in the order shown. Batteries shall meet the requirements of 3.7.12.1.

a. Altitude cycling. After the following altitude cycling test, batteries shall meet the requirements of 3.7.12.1a.
   i. Weigh each battery to the nearest hundredth of a gram and visually inspect all test samples in accordance with paragraph 4.6.1.4. Record weight of each sample.
   ii. Perform battery open-circuit voltage test in accordance with paragraph 4.6.1.5.1 and record voltage.
   iii. Place sample batteries in a test chamber and adjust air temperature to -26±5ºF (−32ºC) and a pressure of not less than 14.7 psia. Batteries shall stay at these conditions for not less than 4 hours.
   iv. Reduce the pressure to not greater than 1.69 psia, the equivalent to an altitude of 50,000 feet above sea level, at a rate not greater than 3.5 feet per second (f/s), while maintaining the temperature conditions of –26±5ºF. Hold these conditions for not less than 1 hour.
   v. Return the pressure to not less than 14.7 psia, the equivalent to an altitude of sea level, at a rate not greater than 3.5 f/s, while maintaining the temperature conditions of -26±5ºF. Hold these conditions for not less than 1 hour.
   vi. Repeat steps iv and v for a total of 10 cycles.
   vii. Adjust the chamber temperature to 70±5ºF (21ºC) in not greater than 1 hour. Hold these conditions for not less than 4 hours.
   viii. Repeat Step 1 and Step 2. In addition to the criteria of paragraph 4.6.1.4, inspect all labels on the battery to include possible damage to any adhesives that may have been used.
b. Explosive decompression. Batteries shall be subjected to the explosive decompression test of MIL-STD-810, Method 500, Procedure IV. Weigh each sample to the nearest hundredth of a gram and record. Connect each battery to a test fixture as specified (see applicable specification sheet). Ramp pressure down at a rate not greater than 3.5 feet per second (f/s) to establish the cabin altitude. After sustaining the low pressure conditions for the time specified in MIL-STD-810, reduce air pressure to the required test altitude of MIL-STD-810 in not greater than 0.1 seconds. Hold these conditions for not less than 10 minutes, then return the test item to ambient conditions at a rate not greater than 3.5 f/s. Visually inspect all test samples in accordance with paragraph 4.6.1.4, perform battery voltage tests in accordance with paragraph 4.6.1.5 and weigh each battery to the nearest hundredth of a gram upon completion of the explosive decompression test of MIL-STD-810. Batteries shall meet the requirements of 3.7.12.1b.
c. Parachute drop. Weigh each sample to the nearest hundredth of a gram and record. Connect each battery to a test fixture as specified (see applicable specification sheet) prior to the drop. Drop each battery once from a height of 10 feet into a synthetic solution containing 3.6 percent sea salt in accordance with ASTM-D-1141 with a depth of not less than three feet. Towel-dry batteries and leave to air dry for not less than 15 minutes after the drop. Visually inspect all test samples in accordance with paragraph 4.6.1.4, perform battery voltage tests in accordance with paragraph 4.6.1.5, and weigh each battery to the nearest hundredth of a gram upon completion of the parachute drop test. Batteries shall meet the requirements of 3.7.12.1c.
d. Submersion. Weigh each sample to the nearest hundredth of a gram and record. Connect each battery to a test fixture as specified (see applicable specification sheet). Submerge test samples in saltwater substitute salt in accordance with ASTM D-1141. Pressurize the

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T05:38Z
Check the source to verify that this is the current version before use.

solution with the battery samples to simulate three-foot submersion for not less than 24 hours and then raise the pressure to simulate 50-foot submersion for not less than 15 minutes. Towel-dry the battery samples and leave to air dry for not less than 15 minutes. Visually inspect all test samples in accordance with paragraph 4.6.1.4, perform battery voltage tests in accordance with paragraph 4.6.1.5, and weigh each battery to the nearest hundredth of a gram upon completion of the submersion test. Batteries shall meet the requirements of 3.7.12.1d.

e. Capacity test H2. The test method of 4.6.4.1 shall be used for the H2 capacity test. Discharge at +130 ±5ºF (55ºC) to cut-off voltage after stabilization storage at +130 ±5ºF for not less than three and one-half (3.5) hours.

Add: The following to TABLE III of MIL-PRF-32271 after “Connector”:

Group No.	Sample Size	Inspection	Requirement Paragraph	Test Method Paragraph
III	23	Magnet test	3.4.11.1	4.6.1.12.1

Add: The following to TABLE IX of MIL-PRF-32271 after “Charge protection”:

Inspection or Test	Requirement Paragraph	Method or Test Paragraph	Test sample break-out by test
Magnet test 1/	3.4.11.1	4.6.1.12.1	All

NOTES:

(This section contains information of a general nature which may be helpful, but is not mandatory)

Intended use: This specification covers non-rechargeable lithium batteries of the non-reserve type composed of electrochemical cells using lithium metal in the anode. They are intended for use in military communications and electronics equipment.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T05:38Z
Check the source to verify that this is the current version before use.

Nominal ratings[5]: The following are the nominal ratings for each battery type described by this specification sheet. They are provided for information purposes.

Battery PIN:	**M32271/10-12D (Li/MnO_2)**
Overall Dimensions:	3.31 in H x 2.25 in W x 1.54 in D
MAX Weight:	0.6 lbs (272 grams)
Voltage Range:	7.5 – 13.20
Nominal Capacity: (in ampere-hours)	3.25 Amp-hrs
Nominal Energy: (in watt-hours)	39 Watt-hrs
MAX rated power output:	26 watts (3s pulse)
MAX continuous load rating:	1.0 ampere
MAX pulse load rating:	3.5 amperes
Instantaneous trip rating:	4.0 amperes
Operating temperature range:	-4 to 130ºF (-20 to 55ºC)
Storage temperature range:	-40 to 160ºF (-40 to 71ºC)
MAX abusive temperature: (non-operating)	195ºF (91ºC)

Other data:

Military Type Designations. The military type designation that relates to the PIN covered by this specification sheet is as follows: The BA-5312()/U designation has been used for the Type I PIN. It should be noted that the Type II BA-5112()/U has been replaced by the BA- 5312()/U.

This battery is subject to the transportation requirements of 49 CFR 173.185. The battery has solid cathodes and more than 2.0 grams of lithium content. See applicable Material Safety Data Sheet (MSDS) for the maximum lithium weight per cell and battery. It should be noted that all non-rechargeable lithium batteries are restricted from shipment as cargo aboard passenger aircraft within, entering, or leaving the US.

This battery includes a complete discharge device in order to render the battery as non-reactive after use. If the battery is damaged, or if the device does not operate as indicated by an amber or yellow light, the battery is considered reactive waste. This battery may be recycled after use. Universal waste rules, where applicable, apply only when recycling is the chosen disposal method.

The US Army CECOM Life Cycle Management Command (LCMC) publishes guidelines for the design of battery compartments for devices using both battery Types I covered by this specification sheet: Technical Bulletin 43-6135. Please contact the CECOM LCMC Directorate for Safety at usarmy.APG.cecom.mbx.amsel-sf-sec-cecom-lcmc--ds@army.mil for further information.

Navy safety tests of NAVSEA S9310-AQ-SAF-010 are required for this battery during first article testing. The test data provided from the testing will be used to evaluate applications for specific using devices and Navy platforms. Devices using this battery will require US Navy Safety Approval prior to use by Department of Navy users unless such approval has already been granted. Please consult with NAVSEA Instruction 9310.1 for further information.

[5] Capacity and energy delivered by a battery are significantly affected by usage conditions, such as temperature and loads applied. If you have any questions about use of this battery in a particular device or circumstance, please visit the following web site (contact info is posted): https://battery.army.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T05:38Z
Check the source to verify that this is the current version before use.

The previous version of this document required “Watertight integrity” testing; however, this test was revised in the latest update. The requirement and verification, previously known as “Watertight integrity,” are now under “Additional environmental requirements and tests.”

Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

Custodians:
Army – CR
Navy – SH
Air Force – 99
DLA – CC

Review activities:
Navy – AS
Air Force – 71

Preparing activity:
Army – CR
(Project Number 6135-2025-011)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil or https://quicksearch.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T05:38Z
Check the source to verify that this is the current version before use.